

Hydroecological degradation of the Kura River through water quality pressures and its implications for ecosystem services and local green economy development

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Keywords

ecosystem services, green economy, hydroecology, Kura River, water quality

ABSTRACT

The Kura River is Azerbaijan's principal surface-water artery and supports drinking-water supply, irrigated agriculture, energy production and aquatic ecosystems. This study integrates two educational research projects that approached hydroecological degradation from complementary scales: a literature-based assessment of Kura River water quality and flow pressures, and a local field-monitoring project in Padar village that examined visible pollution, environmental awareness and a low-cost filtration model. The synthesis was verified against recent peer-reviewed and institutional evidence and interpreted through ecosystem-services and green-economy perspectives. Recent monitoring for 2020–2022 shows a strong downstream mineralization gradient in the Kura, rising from about 349 mg/L at I Shikhli to more than 3,000 mg/L at Neftchala, while the local project documented plastic litter near the village, organic and fertilizer residues near pasture areas, and improved visual clarity after mechanical filtration through sand, gravel and activated carbon. The filter experiment demonstrates physical pre-treatment only and does not establish drinking-water safety. Economically, declining water quality can increase treatment and monitoring needs, constrain irrigation suitability, intensify salinity-related production risks and weaken fisheries, recreation and other ecosystem services. The combined evidence supports prevention-oriented water management, stronger wastewater control, community monitoring and resource-efficient local interventions as components of Azerbaijan's green-economy and water-security agenda.

INTRODUCTION

The Kura River functions simultaneously as an ecological corridor, a source of drinking and irrigation water, a hydropower resource and a foundation for settlements and productive activity in Azerbaijan. This multifunctional role makes water quality an environmental and economic issue rather than a purely chemical one. When pollution, salinity and altered flow reduce the suitability of river water, the consequences extend from aquatic organisms to farms, municipal treatment systems, public health, fisheries and local livelihoods. Hydroecological degradation therefore represents a loss of natural-capital quality as well as a threat to water security.

The two source research works used in this article examine that problem at different scales. Ismayıl's project focused directly on the Kura River and assembled evidence on transboundary pollution, wastewater discharge, flow alteration, mineralization and ecological effects. Karimova's project was carried out in Padar village, Khachmaz, at a local water body that is not a direct Kura tributary. Its relevance is therefore not as a measurement of Kura water quality, but as a community-scale demonstration of pollution pressures, field observation, environmental awareness and practical mitigation within Azerbaijan's wider connected water environment. This distinction is retained throughout the analysis so that local observations are not incorrectly generalized to the Kura basin.

The economic dimension is central to the journal scope. Polluted or highly mineralized water can require additional treatment before municipal use, reduce the flexibility of irrigation, contribute to salinity risks and weaken ecosystem services that support fisheries, recreation, landscape quality and local well-being. These are economic pathways even when a monetary valuation is unavailable. In a green-economy framework, the objective is to maintain or increase social and productive value while reducing pollution loads, material waste and pressure on aquatic natural capital.

The study therefore asks three questions: (1) what water-quality and hydroecological pressures are supported by the combined source materials and recent evidence; (2) what can local field monitoring and low-cost interventions contribute without overstating their scientific scope; and (3) how do water-quality improvements connect to ecosystem services, resource efficiency and local green-economy development?

LITERATURE REVIEW

Recent peer-reviewed monitoring substantially strengthens the water-quality observations contained in the original Kura research. Aliyev and Abduev (2025) analysed seasonal samples collected in 2020–2022 along the Kura in Azerbaijan and found a persistent downstream increase in mineralization and major ions. Mean mineralization increased from approximately 349 mg/L at I Shikhli to

404 mg/L at Mingachevir, 777 mg/L at Sabirabad, 921 mg/L at Salyan and 3,086 mg/L at Neftchala. The same study reports that the lower river changes from calcium-bicarbonate to sodium-sulfate water and links this pattern to tributary inflows, wastewater, mineralized groundwater, the Aras confluence and interaction with Caspian water near the mouth.

The basin context is intrinsically transboundary. UNECE (2011) identifies wastewater discharge, agricultural pressure, hydromorphological modification and inadequate infrastructure as recurrent pressures in the Kura-Aras system. Strosser et al. (2017) further argue that stronger Georgia-Azerbaijan cooperation can generate economic, environmental and social benefits, even though complete monetary valuation is constrained by data gaps. This literature supports the source project's emphasis on upstream-downstream interdependence while avoiding attribution of individual pollution events without traceable monitoring evidence.

Water quality is also connected to the national economy through water-dependent agriculture and infrastructure. The World Bank (2024) notes that Azerbaijan relies heavily on transboundary inflows and that irrigated agriculture is a major water user and an important sector for employment, livelihoods and economic diversification. In this context, poor water quality can create indirect costs through treatment requirements, irrigation limitations, soil degradation and lower resilience to drought. The relevant economic concept is therefore not only the price of water, but the productivity and ecosystem value generated per unit of safe and usable water.

National policy increasingly reflects this linkage. The National Strategy on the Efficient Use of Water Resources for 2024–2040 prioritizes integrated management, efficient use, reliable water-supply and sanitation services, irrigation provision and water security. Azerbaijan 2030 also places efficient resource use and a clean environment within the national green-growth priority. For the Kura River, these objectives imply that pollution prevention, monitoring, wastewater treatment and efficient allocation should be evaluated together rather than as isolated environmental measures.

At the local scale, community monitoring and environmental education can complement formal systems by identifying visible litter hotspots, fertilizer residues and behavioural problems that are often missed by infrequent technical sampling. Such initiatives cannot replace laboratory analysis or regulatory enforcement, but they can reduce local waste leakage, generate environmental awareness and support low-cost preventive action. The Padar village project is interpreted in this limited but useful role.

MATERIALS AND METHODS

The study used an integrative qualitative design based on two author-prepared research dossiers completed in 2026. The first dossier, prepared by Ismayıl, synthesized official monitoring information, dissertation-based hydroecological material and conference sources on the Kura River. It focused on water-quality deterioration, flow change, industrial and municipal wastewater, agricultural pressure, dams and downstream mineralization. The second dossier, prepared by Karimova, documented a school-based field project in Padar village using visual monitoring, simple physical observations, a 15-respondent awareness survey, cartographic review and a classroom filtration model.

Karimova's field team reported monitoring at three locations, but the source results table contains interpretable findings for two points only. Point A, close to the village, was classified by the project team as critically highly polluted because of polyethylene bags and plastic bottles. Point B, near a pasture area, contained organic waste and

fertilizer residues and was classified as moderately polluted, with eutrophication identified as the principal ecological risk. The project also tested a layered mechanical filter made of plastic housing, sand, gravel and activated carbon and reported improved visual clarity of turbid water. No microbiological, chemical or potability test was supplied.

For external verification, the synthesis used Aliyev and Abduev (2025), the World Bank's 2024 water-sector assessment, UNECE basin assessments, the OECD study of transboundary cooperation and Azerbaijan's current water and green-growth policy framework. Quantitative values were included only where they were traceable to these sources. The 15-person survey is acknowledged as part of the original project design, but item-level response frequencies were not available in the dossier and were therefore not statistically analysed. The economic assessment is interpretive and does not estimate treatment costs, farm-income losses or ecosystem-service monetary values.

RESULTS AND DISCUSSION

Downstream water-quality deterioration and hydroecological pressure

The strongest verified quantitative pattern is the downstream rise in mineralization. Aliyev and Abduev (2025) report mean values increasing from about 349 mg/L at I Shikhlı to more than 3,000 mg/L at Neftchala. This confirms the central concern of Ismayıl's original research that the lower Kura faces substantially greater hydrochemical stress than the river near the Georgian border. The pattern is not attributable to a single source: the peer-reviewed study identifies the combined influence of tributaries, wastewater discharges, the Aras River, mineralized groundwater and Caspian mixing near the estuary.

The hydroecological consequence is a progressive narrowing of water-use options. Higher salinity and major-ion concentrations can reduce the suitability of water for some irrigation conditions, intensify soil-salinity management requirements and complicate treatment for municipal use. From an ecosystem perspective, the same river sections are exposed to altered chemical conditions alongside hydrological regulation and habitat fragmentation. The source work also discusses historical reductions in downstream flow and migration barriers associated with dams; these observations are retained qualitatively because the original numerical comparisons were not accompanied by a fully traceable primary dataset.

Figure 1 presents selected mean mineralization values from the 2020–2022 monitoring dataset reported by Aliyev and Abduev (2025). The steep increase below the middle reaches should be interpreted as a water-quality gradient, not as a direct measure of toxicity or economic loss.

Table 1 combines the Padar village field evidence with the intervention evidence from the classroom filter trial. The purpose is to preserve the original project observations while separating visible pollution indicators from laboratory water-quality claims.

Local pollution observations, environmental awareness and low-cost filtration

The Padar project identifies two common local pathways by which human activity can degrade small water bodies: plastic litter near settlements and nutrient-rich organic or fertilizer residues near pasture areas. These observations are site-specific, but the mechanisms are consistent with wider water-management literature. Plastics create persistent physical pollution and can fragment into smaller particles, while nutrient inputs can stimulate eutrophication and oxygen stress. The project's value lies in showing how visible pollution can be

classified through structured school-based monitoring and translated into cleanup and awareness actions.

The 15-respondent survey demonstrates an attempt to include community perceptions, but the absence of item-level frequencies prevents quantitative interpretation. Similarly, the low-cost filter improved visual clarity, which is a meaningful demonstration of sediment and particle removal, but it cannot be treated as a drinking-water treatment result. Activated carbon, sand and gravel may improve physical appearance and adsorb some compounds under controlled conditions, yet safe water requires validated treatment, disinfection and chemical and microbiological testing. This distinction is important for responsible environmental education.

Table 1. Field observations and practical intervention evidence from the Padar village project

Evidence / observation	Hydroecological and green-economy interpretation
Padar point A – polyethylene bags and plastic bottles; project classification: critically high visible pollution	Persistent litter pressure; habitat and amenity degradation; recurring cleanup burden and material leakage
Padar point B – organic waste and fertilizer residues; project classification: moderate	Nutrient-loading and eutrophication risk; need for manure and fertilizer management near water bodies
Classroom layered filter – sand, gravel and activated carbon	Visual clarity improved; useful as physical pre-treatment demonstration, but not evidence of potable water
Lower Kura – strong downstream mineralization gradient in published 2020–2022 monitoring	Greater treatment and irrigation-management pressure; risk to ecosystem services and local economic resilience

Water-quality degradation as an economic and ecosystem-services issue

Hydroecological degradation generates economic exposure through several connected channels. First, lower raw-water quality can increase monitoring and treatment requirements before municipal use. Second, higher salinity and mineralization can reduce irrigation flexibility and contribute to soil-management costs in already water-stressed agricultural areas. Third, degraded aquatic habitat can weaken fisheries and other provisioning services. Fourth, littered or polluted riverbanks reduce recreational and landscape value. These pathways do not prove a specific monetary loss in the present study, but they show why water-quality protection is an investment in productive and social assets rather than an environmental expense alone.

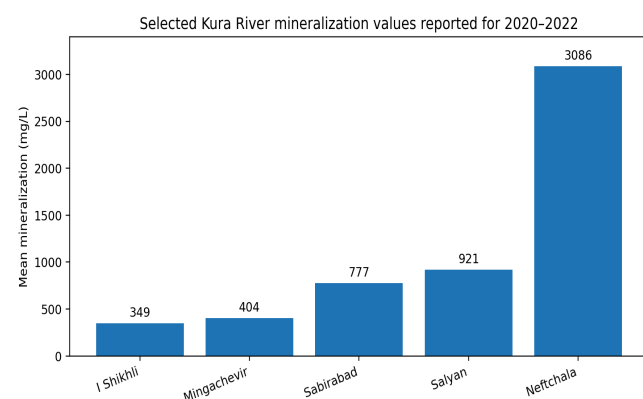


Figure 1. Selected mean mineralization values along the Kura River reported by Aliyev and Abduev (2025) for 2020–2022

The sharp lower-river increase is particularly relevant to green-economy planning because natural capital becomes less

productive when more resources are needed to make water usable or when land, fisheries and recreation lose ecological quality. Prevention can therefore have a double return: avoiding degradation costs while preserving ecosystem services. This logic is consistent with the World Bank's recommendation to strengthen water-quality monitoring, water accounting and the valuation of water as an economic resource within integrated management.

Green-economy responses from basin infrastructure to community action

The combined evidence supports a layered response. At basin and municipal level, priorities include reliable wastewater collection and treatment, monitoring of tributaries and industrial discharges, transparent water-quality data, pollution-control enforcement and coordinated transboundary management. At agricultural level, nutrient management and efficient irrigation can reduce return-flow pollution and pressure on the river. At community level, litter prevention, organized shoreline cleanup, school monitoring and bilingual awareness materials can address visible local pressures. Low-cost filters can be used for education and pre-treatment demonstrations, but not as substitutes for regulated drinking-water systems.

Local green economy development, governance and study limitations

Local green-economy development can build on water stewardship without requiring large industrial projects. Cleaner riverbanks and reliable water quality support recreation, small-scale nature tourism, fisheries-related livelihoods, greener public spaces and lower cleanup burdens. School and community projects can also strengthen environmental skills and create demand for waste sorting, recycling, monitoring services and resource-efficient technologies. These benefits become more durable when local initiatives are connected to municipal services and national water policy rather than remaining one-off campaigns.

The study has important limitations. The two underlying works are educational research projects rather than standardized hydrochemical surveys. The Padar field site is not a Kura monitoring station and should not be used to infer Kura water quality. The source file lists three field points but reports interpretable results for only two; survey response distributions are unavailable; and the filtration experiment lacks chemical and microbiological measurements. The article therefore uses verified published data for quantitative Kura claims and treats the student observations as descriptive local evidence. Future work should add georeferenced sampling, repeated seasonal measurements, basic physicochemical parameters, laboratory testing and cost-benefit assessment of treatment and pollution-prevention options.

CONCLUSIONS

This study integrates a Kura-focused hydroecological review with a local field-monitoring project to show how water-quality degradation connects environmental condition, ecosystem services and green-economy resilience. Verified 2020–2022 evidence confirms a pronounced downstream mineralization gradient, with mean mineralization rising from about 349 mg/L at I Shikhli to more than 3,000 mg/L at Neftchala. The local Padar project independently demonstrates visible plastic and nutrient-related pollution pressures and shows the educational value of structured monitoring and simple filtration experiments. However, the filter result is limited to visual clarification and the local field observations cannot be generalized to the Kura River. The practical implication is that effective protection requires action at multiple levels: wastewater control and basin

monitoring, efficient agricultural practices, transboundary coordination, local litter prevention and environmental education. From a green-economy perspective, such measures protect natural capital by maintaining usable water, reducing avoidable treatment and cleanup pressures and sustaining agriculture, fisheries, recreation and community well-being. Future research should combine school and citizen-science engagement with standardized laboratory monitoring and explicit economic valuation of water-quality improvements.

AUTHOR CONTRIBUTIONS

L.I. contributed the original Kura River hydroecological research material, source collection and preliminary synthesis. G.K. contributed the local field-monitoring project, pollution classification, survey design, practical filtration model and project documentation. G.G. contributed supervision, conceptual integration, methodological framing, economic and green-economy interpretation, review and editing. All authors reviewed the final manuscript and approved the submitted version.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

FUNDING

This research received no external funding.

DATA AVAILABILITY

This study synthesizes two author-prepared educational research dossiers and publicly available secondary sources. No new laboratory water-quality dataset or economic valuation dataset was generated. The Padar project

reported descriptive field observations and a 15-respondent survey, but item-level survey frequencies and laboratory measurements were not included in the source dossier. Quantitative Kura River values used in this article are taken from the cited published sources.

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